

Volatility-Based Trading Strategy

Algorithmic Trading

Proprietary Research | Subject to Further Review and Evaluation

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Overview

What is an Options Straddle

Straddle Technique

How does it work?

- Simultaneous Buy: A straddle involves simultaneously buying both a call option and a put option on the same underlying asset, with the same strike price and expiration date.
- Betting on Volatility: The strategy profits when the underlying asset's price makes a significant move (either up or down) beyond the combined premiums paid for the call and put options. It's a bet on increased volatility.
- Limited Loss, Unlimited (or Large) Potential Gain: The maximum loss is the total premium paid for both options. The potential profit is theoretically unlimited on the upside (call) and substantial on the downside (put), limited only by the asset price reaching zero.

Black Scholes Model

$$C = SN(d_1) - Ke^{-rT}N(d_2)$$

$$d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}} \text{ and } d_2 = d_1 - \sigma\sqrt{T}$$

Required Inputs:

S = Current stock price

K = Option strike price

r = Risk-free interest rate

T = Time remaining until option expiration

σ = Volatility of the stock

Straddle Graph

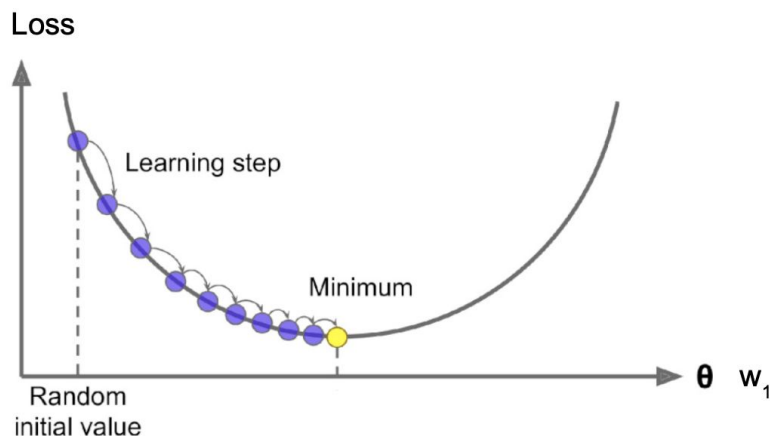
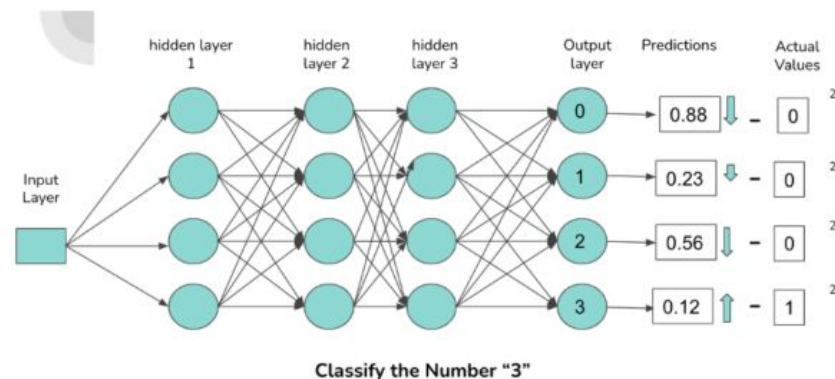


Neural Networks

Gradient Descent

Definitions

- A brain-inspired way to process data
- Each node takes multiple input values, adds them up, and converts the output into an on-or-off signal
- Outputs of all nodes in one layer are then passed into the next layer
- Each node has an activation function, which based on the value of the inputs each node either activates or doesn't.



Gradient Descent

- Gradient descent is the method of training the model
- Core optimization algorithm used in neural networks to train the model's parameters (activation function).
- The goal is to find the "place" (set of parameters) with the lowest loss, by gradually adjusting the parameters.

Indicators

Realized (Historical) Volatility

Realized Volatility

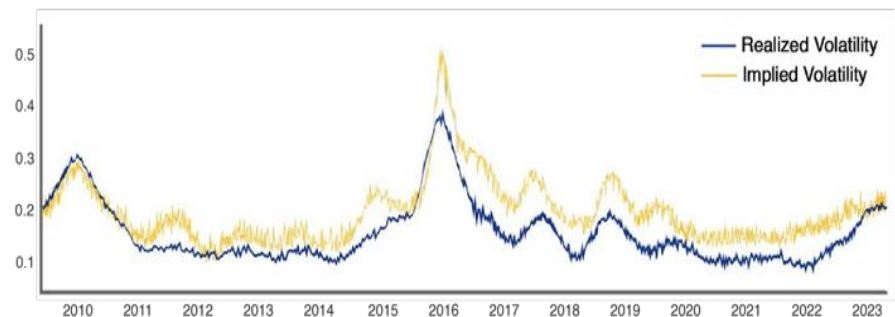
• What does Realized Volatility measure?

- Past Price Fluctuations: Realized volatility measures the actual fluctuations of an asset's price over a specific historical period.
- Statistical Calculation: It's calculated using historical price data, typically as the standard deviation of returns, providing a quantifiable measure of past price variability.
- Backward-Looking Indicator: Unlike implied volatility, which is forward-looking, realized volatility is a historical measure, reflecting what has happened, not what is expected to happen.

• Components

$$Vol = 100 \cdot \sqrt{\frac{252}{n} \sum_{t=1}^n R_t^2}$$

• Example Graph



Average True Range (ATR)

ATR

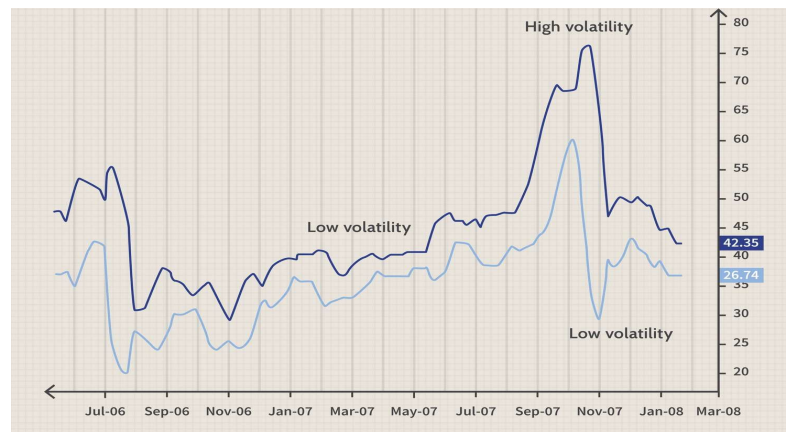
• What does ATR measure?

- Volatility Measure: ATR indicates the average price fluctuation of an asset over a specified period, capturing both typical and gap-related volatility.
- True Range Calculation: It calculates the "true range" as the greatest of the current high minus the current low, the absolute value of the current high minus the previous close, or the absolute value of the current low minus the previous close.
- Non-Directional Indicator: ATR measures volatility, not price direction, and is used to assess potential price swings and inform stop-loss placement or position sizing.

• Components

$$TR = \max[(\text{high} - \text{low}), \\ \text{abs}(\text{high} - \text{close}_{\text{prev}}), \\ \text{abs}(\text{low} - \text{close}_{\text{prev}})]$$
$$ATR = \frac{1}{n} \sum_{i=1}^n TR_i$$

• Example Graph



IQR Volatility

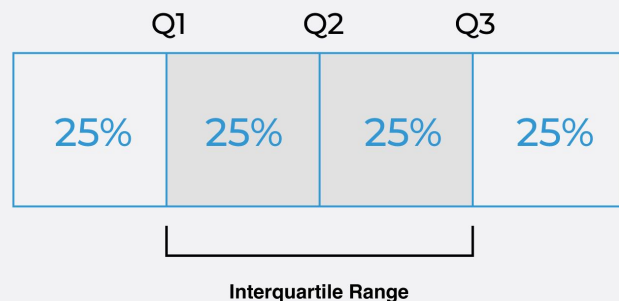
IQR

• What do Stochastic Oscillators measure?

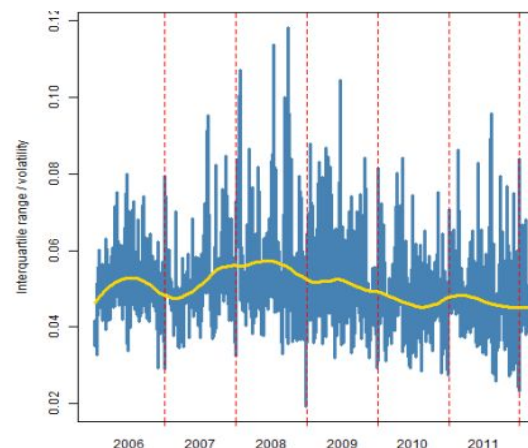
- Measures Central Price Dispersion: The IQR represents the range between the 25th percentile (Q1) and the 75th percentile (Q3) of price data, capturing the volatility of the middle 50% of observations.
- Robust to Outliers: Unlike standard deviation which is sensitive to extreme values, the IQR is a more robust measure of volatility as it ignores the highest and lowest 25% of prices.
- Typical Price Swings: A larger IQR suggests greater typical price fluctuation within the central portion of the data, while a smaller IQR indicates less volatility in the core price movements.

• Components

Visualizing the Interquartile Range (IQR)



• Example Graph



Bollinger Spread

Bollinger Bands

• What does Bollinger Spread measure?

- Normalized Bandwidth: Bollinger Spread normalizes the width of the Bollinger Bands by dividing it by the middle band (simple moving average). This provides a relative measure of volatility.
- Facilitates Cross-Asset Comparison: By normalizing, Bollinger Spread allows for a more direct comparison of volatility levels across different assets with varying price scales.
- Identifies Relative Highs and Lows in Volatility: High Bollinger Spread readings indicate relatively high volatility compared to the asset's recent history, while low readings suggest relatively low volatility.

• Components

Upper band = 20-day SMA + (20-day SD x 2)

Middle band = 20-day SMA

Lower band = 20-day SMA – (20-day SD x 2)

Spread = Upper-Lower = 4 x 20-day SD

• Example Graph



Rogers Satchell Volatility

Rogers-Satchell

• What do Stochastic Oscillators measure?

- Range-Based Estimator: Rogers-Satchell is a range-based volatility estimator that considers the high, low, open, and close prices of an asset, making it more efficient than estimators relying solely on closing prices.
- Handles Gaps and Drifts: It is designed to provide a more accurate volatility estimate, particularly in the presence of price jumps (gaps) and non-zero drift (directional movement) in the asset's price.
- More Efficient Than Historical Volatility: By incorporating intraday high and low prices, it can provide a more precise and potentially faster indication of volatility compared to traditional historical volatility calculated from closing prices alone.

• Components

ROGERS-SATCHELL VOLATILITY FORMULA

$$\sigma_{RS} = \sqrt{\frac{1}{T} \sum_{t=1}^T \left(\ln\left(\frac{h_t}{c_t}\right) \ln\left(\frac{h_t}{o_t}\right) + \ln\left(\frac{l_t}{c_t}\right) \ln\left(\frac{l_t}{o_t}\right) \right)}$$

Where:

T — Number of days in the sample period

o_t — Open price on day t l_t — Low price on day t
 h_t — High price on day t c_t — Close price on day t

• Example Graph



Chaikin Volatility

Chaikin

• What does Chaikin Volatility measure?

- Measures Range Expansion: Chaikin Volatility quantifies volatility by analyzing the difference between two moving averages at different times. It highlights periods of range expansion (increasing volatility) and contraction (decreasing volatility).
- Essentially, this involves calculating the exponential moving average (which weighs the more recent prices more) of high and low prices of a stock over a period of 10 days (recommended by Chaikin).
- Identifies Potential Trend Changes: Sharp increases in Chaikin Volatility can signal the beginning of a new trend, while decreasing volatility might suggest trend exhaustion or consolidation.

• Components

High Low Average = EMA of (High – Low)

Volatility = $\frac{[(\text{High Low Average} - \text{High Low Average } n \text{ Periods ago}) / \text{High Low Average } n \text{ Periods ago}]}{1} * 100$

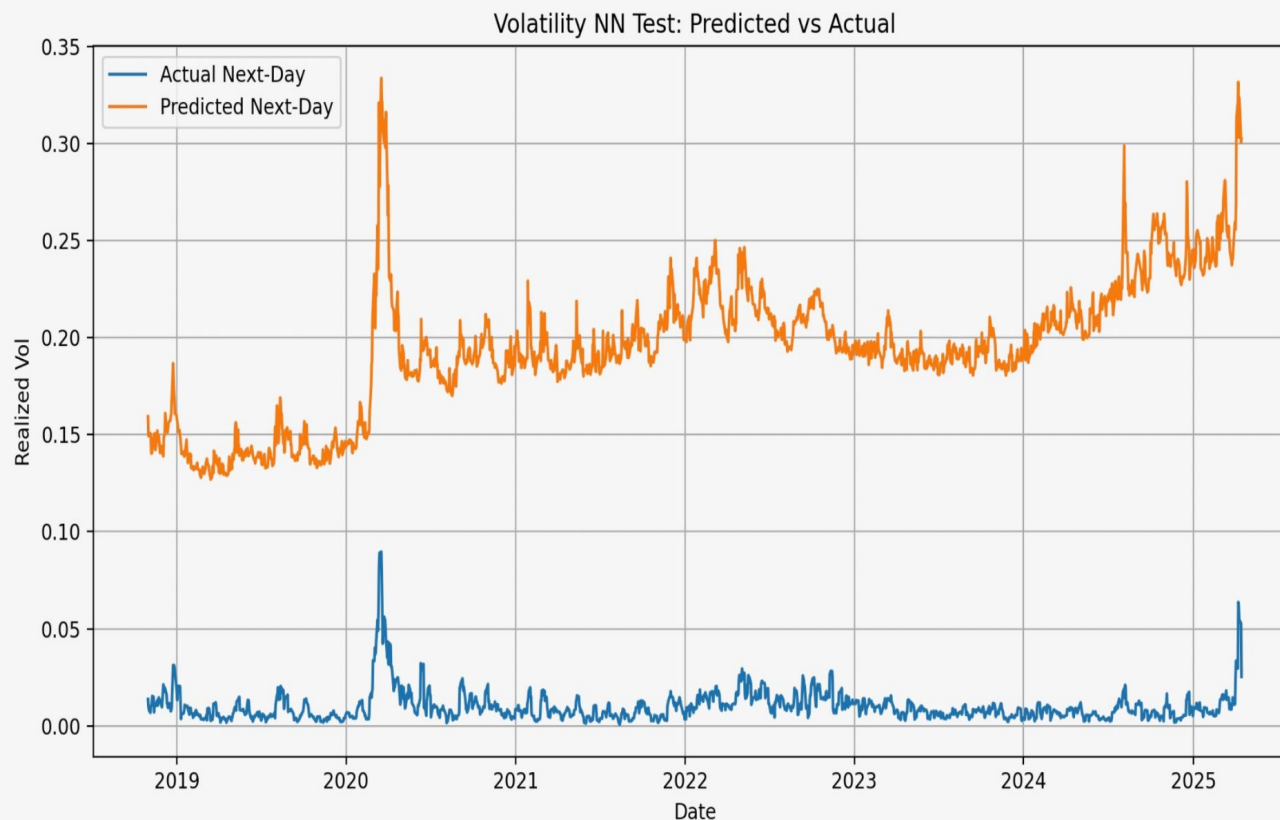
• Example Graph



Application

Testing

S&P 500



Hyperparameters:

Indicator layers:

- 400 epochs
- 1 Hidden Layer
- 128 neurons

Big Model:

- 1000 epochs
- 2 Hidden Layers
- 64 neurons

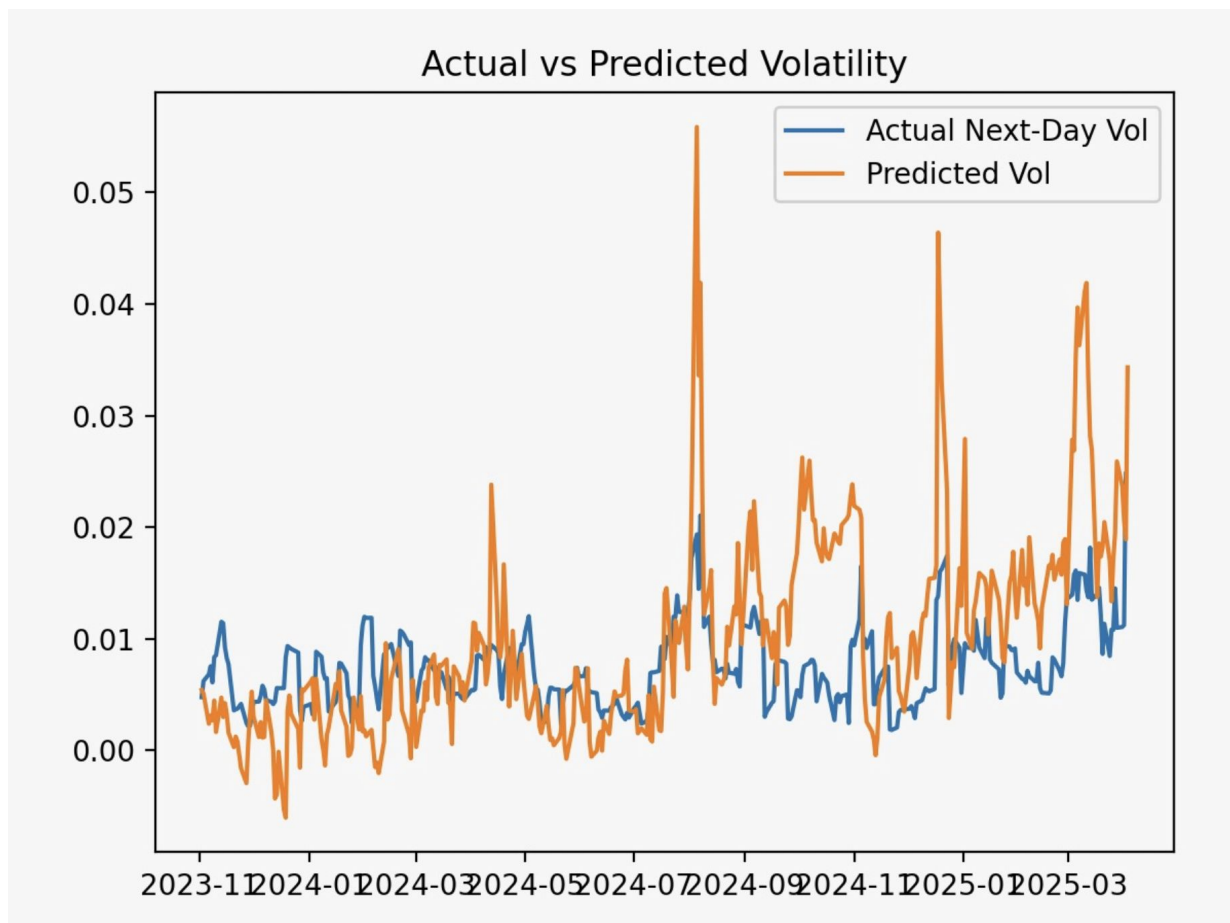
Results:

MAE: 0.1976

**MAE adjusted:
0.0321**

Testing

S&P 500



Cointegration Testing

Engle-Granger Cointegration Test

- To reduce computing times, we want to find cointegrated stocks in order to make sure that we can run the model daily in an efficient manner.
- Cointegration describes a long-term, stable relationship between two or more variables over time, even if these variables are individually non-stationary.
- This relationship is ideal when trying to predict stock volatility in pairs trading, as the volatilities will expect to converge after diversion
- The augmented Engle-Granger two-step cointegration test assumes that the series are non-stationary and integrated to an order of 1.
- The null-hypothesis is that the stocks are not cointegrated.

Two Steps:

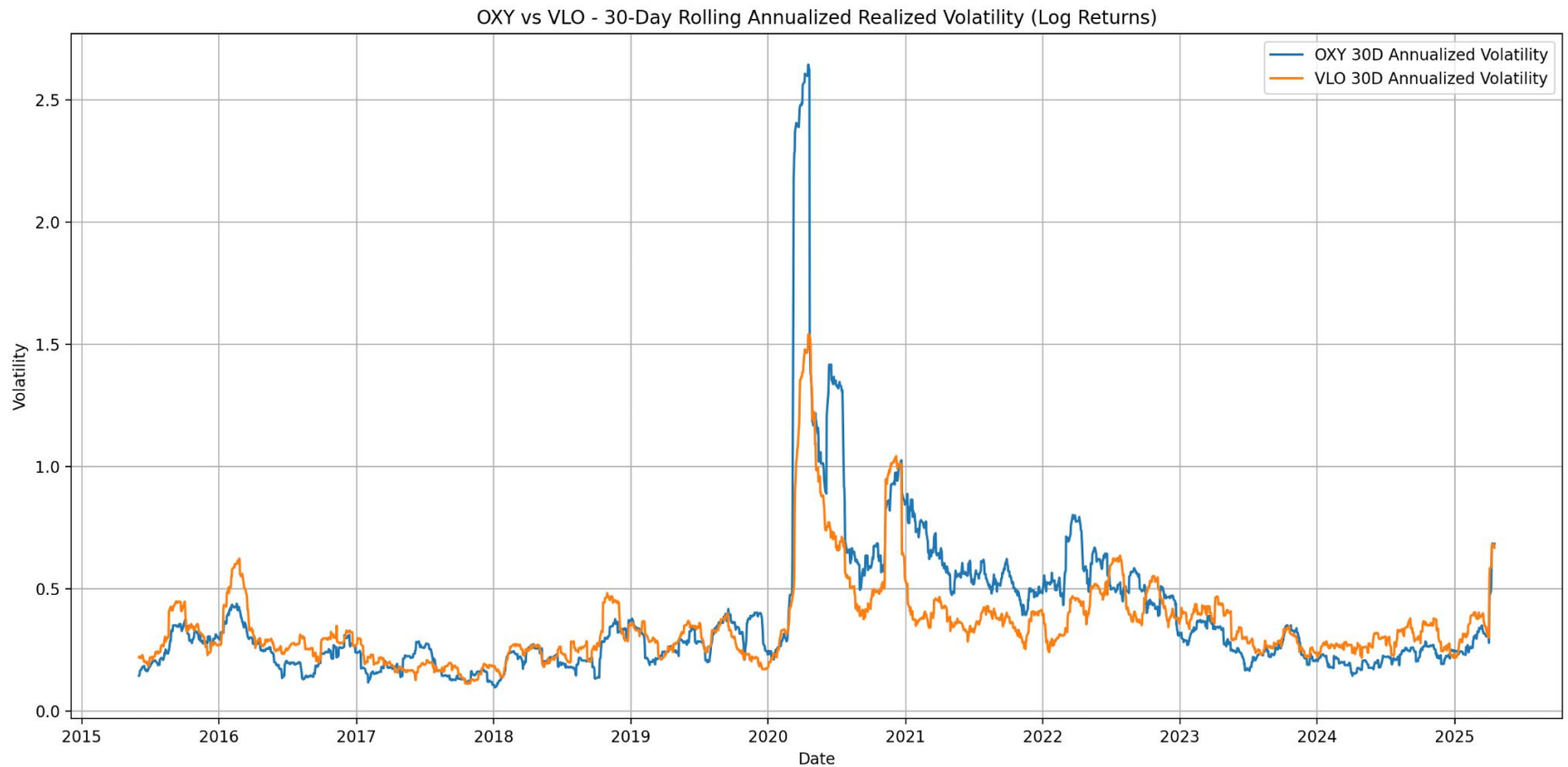
1. Test for Stationarity of Linear Comb.
$$y_t - \beta x_t = u_t$$
 - Estimate through OLS, then run an stationarity test on u_t
2. Regression on the residuals
 - α should not be 0
$$\Delta y_t = \Delta x_t b + \alpha u_{t-1} + \varepsilon_t$$

T-statistics

- Using the t-statistic on the regression of residuals, the certainty of rejection of the null-hypothesis can be assessed.
- The t-statistic value is compared to the critical values for each stock pair at the 95% level of certainty (must be more negative than the critical value).

Cointegration Testing

Case: Occidental Petroleum Corp vs Valero Energy Corp



Strategy & Next steps

Volatility

- Give the historical stock data for every day, calculate every indicators value for that day.
- Use that number along with price that day to get a value for price using a neural network built on top of each indicator
- Feed the mean, std of the price distribution between each indicators prediction to a larger model along with VIX to get a final value for volatility.
- Use this value to get 'fair price' for an options contract and use a long straddle

Trading strategy

- Whenever the model is predicting a volatility increase of greater than 1 standard deviation of current volatility we enter a long straddle.
- Look at cointegrated stocks to create larger bets
- exercise the contract which is giving a profit when making profit of 1 standard deviation.

Next Steps

- Normalising the graph: add a transformation so graph matches the numbers as well.
- Optimising for look forward period
- Accessing options data to get values historical options data.
- Optimising bet size after getting access to options data



Algory Capital

Q&A

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